

Chigchi Bai*

On uncancelledness of events: a view from Mongolian

<https://doi.org/10.1515/tlr-2025-0006>

Received April 30, 2025; accepted November 12, 2025; published online November 21, 2025

Abstract: Once an event occurs, it can never be cancelled by its very nature. This paper, based on examination of the aspectual and modal properties of the completion marker -čih in Mongolian, shows that uncancelledness, a hitherto undocumented aspect of semantics, plays significant roles in aspectual and modal properties of verbs. It is argued that uncancelledness is an semantic primitive, which leads to the thorough completion of (the commencement of) events as well as a resultant state and causes modal effect by the virtue of the information available in the presupposition, say, contradiction.

Keywords: uncancelledness; events; Mongolian

1 Introduction

Once an event takes place, it can never be cancelled by nature and brings about at least two types of post-event states. Suppose that you drank a bottle of water. No one can undo the drinking event because it took place on the temporal axis, which stretches onward, not backward. The drinking event is now in a state of having taken place and this state will never cease because the occurrence of the event will never be cancelled. This type of state is called “R-state” (Parson 1990). While R-state, which uncancelledness of events gives rise to, is characterized in the literature of event semantics, uncancelledness has not. Uncancelledness, however, is supposed to play crucial roles in event semantics, being associated with not only aspectuality but also modality. Suppose that you find the water you drank poisoned. You regret that you drank it and wish that the drinking event had not occurred. However, there is no way

This work was supported by NSSFC [grant number. 21XYY018]. The work reported in this article has been partially presented at the 2024 Annual Conference of Association for Mongolian Language of China. I thank the audience for their comments and questions. I have also benefited from comments by two anonymous reviewers. All errors, if any, are mine.

***Corresponding author: Chigchi Bai**, School of Foreign Languages, Inner Mongolia University of Finance and Economics, Hohhot, China, E-mail: chigchi.bai@imufe.edu.cn. <https://orcid.org/0009-0000-5371-6537>

to undo the event. The R-state resulting from the drinking event remains contradictory to your wish and you are to sigh unhappily that you drank the poisoned water unfortunately. But how would you express your discontented feelings linguistically? Is there a grammatical form to express such discontent? This is supposed to depend on grammatical parameters. Importantly, the speaker's discontented emotion about an irrecoverable situation – R-state resulting from the occurrence of an event – is released by stating emphatically that the event was fully completed and cannot be cancelled. Thus, uncanceledness of events induces the speaker's intensification of their thorough completion, and hence the release of her or his emotion. Bybee et al. (1994: 57–61) reported a number of languages in which some grammatical forms are correlated with (thorough) completion of events with an emphatic value. Although uncanceledness has not been taken into serious consideration in the literature, it, directly related to thorough completion, has a morpho-semantic realm. For example, speakers of Mongolian, as will be illustrated in this paper, use the verbal suffix *-čih* to emphasize that the event was completed thoroughly and cannot be cancelled, which therefore derives aspectual and modal effects in utterances. The following sections elaborate on the nature of this suffix and its relevance to uncanceledness.

After outlining the aspectual and modal meanings of *-čih* in Section 2, I will discuss how to derive its modal meanings and aspectual meanings in Section 3. The core proposal is that uncanceledness is an aspectual primitive, which is the very basic semantic segment that serves to produce aspectual and modal concepts, and that the proper function of *-čih* is to mark (thorough) completion of events, which relates to uncanceledness of events under an instruction such as *“To say that an action can never be cancelled is to say that it is ‘completed thoroughly’”*. In Section 4, I elaborate on the proposal by conducting a formal semantic analysis of the correlation among uncanceledness, R-state and the perfect.

2 Basic properties of *-čih* in Mongolian

The suffix *-čih* in Mongolian is labeled “Бүрмөсөн үйлдэх байдал” (Buhjirgal and Enghe 1977: 240; Nasunbayer et al. 1982: 358; Haserdeni et al. 1996: 476, and others), literally meaning “aspect of thorough performance”, by traditional grammarians. However, what exactly the so-called ‘thorough performance’ (or ‘thorough completion’) of an action is and how it relates to the modal properties of the suffix have not been explicated in a principled way. Neither has the connection of the suffix with general linguistic theories been made.

2.1 Syntactic properties of *-čih*

We begin by overviewing syntactic and semantic properties of *-čih*, which altogether point to its role in revealing the basic notion of uncanceledness. Four syntactic properties of *-čih* are noted. First, it follows voice suffixes, for example, the passive *-gd (-t)* and the causative *-uul*, and precedes tense and (sentential) aspect suffixes, for example, *-san* and *-dag*.

- (1) Hamag honi-oo conon-d id-üül-čih-sen.
all sheep-RX wolf-DAT eat-CS-čih-PST¹
'(Someone) had all his/her sheep eaten by a wolf.'
(http://www.mongoltoli.mn/search.php?ug_id=105028&opt=1&rating=down)
- (2) Niislel hoto-d boroo orhoor yabgan хүний зам usan-d
capital city-DAT when it rains sidewalk water-DAT
ab-t-čih-dag.
take-PS-čih-HAB
When it rains in the capital city, the sidewalks are flooded.
(<https://www.dorgio.mn/p/54355>)

Second, it follows verbal aspect suffixes,² for example, *-lz*, and precedes sentential aspect morphemes, for example, *-dag* and *-aad.bai*.

- (3) Örgön heregleenii baraanii үнэ helbe-lz-čih geed bai-dag.
price of consumer goods fluctuate-čih usually about to
'The price of consumer goods in the market tends to fluctuate.'
(<https://medium.com/@bolor1931/бадамдоржийн-пунсалмаа-a6d9724a38c8>)

¹ The abbreviations to use in this paper include ACC: accusative, CONJ: conjunction, CS: causative, CV: converb, DAT: dative, GEN: genitive, HAB: habitive, INF: infinitive, MD: modal particle, NOM: nominative, NEG: negative, NPST: non-past, PS: passive, PCL: particle, PROG: progressive, PST: past, Q: question particle, RES: resultative, RX: reflexive-possessive, and VOL: volutive.

² "Verbal aspect" here refers to "inner aspect" (or "lexical aspect", "Aktionsart") and "sentential aspect" refers to "outer aspect" (or "grammatical aspect"). I chose to use "verbal aspect" as a term because in Mongolian, what I call verbal aspect may or may not have over markers such as *-lz*, *-(ga)lz*, *-(ba)lz*, *-s.hii*, *-b.hii* and *-gana*, and all these markers display derivational properties more than inflectional properties. Inflectional properties are displayed by what I call sentential aspect markers such as *-dag*, *-ž.bai*, *-aad.bai* and sometimes *-na*. That *-čih* is sandwiched between two levels of aspect morphemes suggests that it is an aspect marker with an in-between status.

- (4) Cagdaa nar hüühed ala-čih-aad.bai-na.
 policemen child kill-čih-RES-NPST
 ‘The policemen have killed a child.’
 (<https://x.com/ShinebayarTny/status/1816348430677205454?mx=2>)

Third, it is not used in progressive sentences. The progressive form of a verb in Mongolian is created by combining the bare verb stem and the morpheme *-ž.bai*, where the participant is understood as being in the middle of the event. *-čih* can cooccur with *-ž.bai*. However, the resulting string *-čih-ž.bai* does not contribute a progressive meaning. For example, in (5), it cannot be the case that the subject is in the middle of killing her/his child. The verb *al-čih-ž.bai-na* denotes a completed killing event, which is stated from the current viewpoint, as indicated by the present tense auxiliary *bai-na*.

- (5) Hüühed-ee heden cag sabaa-gaar zod-soor bai-gaad
 child-RX for several hour stick-INS beat-PROG be-CV
 al-čih-ž.bai-na.
 kill-čih-RES-NPST
 ‘(Someone) beat her/his child with a stick for several hours and killed it.’
 (https://www.facebook.com/permalink.php/?story_fbid=3784924818192886&id=1652412208110835&locale=ka_GE)

Fourth, it is not used with negated predicates. However, it cooccurs in interrogative suggestions such as *tailburila-čih-güi üü?* ‘Won’t you explain?’, in which *güi* is not a negative marker in the strict sense.

- (6) a. *id-čih-sen.güi.
 eat-čih-PST.NEG
 ‘Intended meaning: did not eat’
 b. *id-čih-ee.güi.
 eat-čih-NPST.NEG
 ‘Intended meaning: have not eaten’

The first two properties indicate that *-čih* is an aspect marker and the last two suggest that it marks completion of events.

2.2 Aspectual properties of *-čih*

Four aspectual meanings of *-čih* are noted. **First**, *-čih* intensifies the completion of the commencement of an event, not of the entire event itself, as exemplified in (7), in which the waiting event has started but has not ended. In the same context, *hülee-sen*, without *-čih*, cannot express the completion of the commencement of the waiting event.

- (7) Hūlee-čih-sen yum čin, hūlee-h I hereg.
 wait-čih-PST CONJ wait-INF have no choice but³
 ‘Since I started waiting, I have no choice but to wait.’

This use of *-čih* is restricted to durative verbs like *hūlee* ‘wait for’ and stative verbs like *sana* ‘miss’ normally.⁴ When used with a stative verb, which does not express a change-of-state meaning, *-čih* brings about a change-of-state interpretation. For example, in (8) the affixed verb *sana-čih-žee* expresses that I got into a state of missing my home after I started doing so. *Sana-žee*, without *-čih*, has no connotation of change-of-state; it simply expresses that the missing event took place in the past.

- (8) Ger-ee sana-čih-žee.
 home-RX miss-čih-PST
 ‘I got homesick now.’

Second, *-čih* intensifies the completion of events. In (9), the action of the matter’s passing away has been completed, where the speaker is suggesting that the past action is not worth mentioning now. Speakers would not use the predicate *öngöreed gar-san*, which does not contain *-čih*, with the same intention since the absence of *-čih* makes the connection between the passing away of the matter and (un)necessity of mentioning it less tight.

- (9) Öngöreed gar-čih-san hereg-ig dahin dahin hel-eed
 pass away-čih-PST matter-ACC again and again say-CV
 yamar tus bai-na daa.
 what benefit be-NPST MD
 ‘It is useless to mention again and again a matter that has passed away.’
 (Chingeltei 1997: 508)

This use of *-čih* has been noted by the traditional grammarians, who label it “Бүрмөсөн үйлдэх байдал”, literally meaning “aspect of thorough performance”.⁵

³ The glossing conventions in this paper are as follows. Glossing to adjuncts and idiomatic expressions is basically phrase-for-phrase, without specifying the internal structure of the phrase. Glossing to particles and conjunctions is basically word-for-word. Glossing to morphemes (only overt endings including verbal affixes and non-nominative case endings are glossed) is morpheme-by-morpheme. Glosses in cited examples are adjusted according to the present conventions.

⁴ When it is used with a punctual verb under this meaning, the verb is forced to denote a repetitive event comprised of multiple single events, where *-čih* expresses the completion of the instance of the events. See discussion on (12).

⁵ Discussion of *-čih* as a completive aspect marker is found in Daobu (1983: 47), Chingeltei (1991: 322, 1997: 507–509), Kullmann and Tserenpil (1996: 133), Chuluu (1998: 178–180), Shiotani (1991: 27), Janhunen (2003: 168, 2012: 146–147), Tegshiduureng (2005: 94–95), Baatarsukh (2009: 70), and Binnick (2012: 144, fn.25).

Note that *-čih* can intensify the completion of an event's commencement in addition to the completion of an entire event, which has not been noticed in the literature. However, in most cases, *-čih* expresses the completion of an entire event, where a non-durative and non-stative verb is preferred. In any case, *-čih* is used to intensify completion/termination.

Third, *-čih* expresses the holistic consumption of the patient. In (10), in which the verb *unš-san* does not contain *-čih*, the book is interpreted as remaining unfinished. As discussed by Otgonsüren (1982), *unš-čih-san*, with *-čih* present, is not appropriate since it entails that the book was finished.

- (10) Bi en nom-ig unš-san, tediil sonirholtai biš
 I this book-ACC read-PST that much interesting not
 bolhoor duusga-ž čadalgüi orhi-san.
 since finish-CV cannot leave it-PST
 'I read this book, but it did not seem interesting and (Otgonsüren 1982: 123)
 therefore I could not finish it, leaving it as it is.'

Similarly in (11), the object is necessarily totally consumed. If you have not finished all your homework, you cannot say *hii-čih-žee* '(I) have done it all'. This is evidenced by the fact that when the statement in the first clause is followed by the negative clause *hii-ž bar-san güi* '(I) did not finish it', there is a contradiction between asserted propositions, leading to the felicity of the sentence. Without *-čih* in the verb, it is not necessarily the case that all of the homework has been finished, and therefore the negative clause is allowed.

- (11) #Bi dasgaal-aa hii-čih-žee, getel bi hii-ž bar-san-güi.
 I homework-RX do-čih-PST however I do-CV finish-PST-NEG
 'I have done all my homework, but I did not finish it.'

The same holds true for the subjects of inherently inchoative verbs like *hail* 'melt (intr)' and *šat* 'burn (intr)', which are affixed by *-čih*, only when the subjects are holistically consumed. All this indicates that *-čih* assigns telicity to the verb, making the event temporally bounded, quite similar to the case of English aspectual particles such as *up* in *eat up*.⁶

Fourth, *-čih* expresses a currently holding state resulting from a prior event. Such a result state (corresponding to Parson's (1990) target state) is sensitive to the lexical semantic type of verbs. As exemplified below, in interrogatives, the locative

⁶ For relevant discussion, see Poppe (1960: 101), Street (1963: 83) and Hangin (1968: 99), who label *-čih* "perfective".

pronoun *haa* ‘where’ usually does not co-occur with *-čih*, if the verb does not entail a present result state. In (12), *haa* does not relate to a state holding at present while in (13), the subject is currently located where the going event ended up, which ensures the current relevance required.

(12) #Ter haa id-čih-sen be?
 he where eat-čih-PST Q
 ‘Where has he eaten?’

(13) Ter haa yab-čih-sen be?
 he where go-čih-PST Q
 ‘Where has he gone?’

Interestingly, an action that remains ongoing, even though its commencement has been completed, also qualifies as a currently holding state, which is expressed by *-čih*. Notice that in this case, the current state is a post-commencement one, not a post-event one. Sentences like (7) and (8) are a clear exemplification of this point.⁷

2.3 Modal properties of *-čih*

In addition to aspectual meanings, certain modal meanings are expressed by *-čih*. While modality has lacked a unified definition, this paper takes it as a category that expresses the speaker’s attitude, emotion and wish towards the proposition, partially following Lyons (1977: 452), rather than a category that exclusively expresses the concepts of necessity, possibility and obligation. This is illustrated by (14) below, in which the modal verb *must* expresses the speaker’s wish rather than an obligation of the hearer (Narrog 2003: 171).

(14) You must tell me how to get to it.

The sundry modal meanings expressed by *-čih* are mostly related with contrary-to-expectation,⁸ where the asserted proposition, which is always affirmative,

⁷ See also Kullmann and Tserenpil (1996: 133), Nelson et al. (1998: 127), Chuluu (1998: 178), and Shiotani (2007: 176), who discuss *-čih* as a resultative marker.

⁸ A reviewer suggests that the contrary-to-expectation property of *-čih* be better being analyzed in terms of mirativity rather than modality. Admittedly, the contrary-to-expectation property is a mirative property. However, *-čih* encodes not only a counterexpectation of the speaker but also the speaker’s attitude towards the proposition, as will be seen in the coming sections. A separate paper will discuss *-čih* in terms of mirativity, which could be regarded as a subclass of modality.

contradicts or diverges from a presupposed proposition, which is always negative. Three among the sundry modal meanings are noted here.⁹ **First**, *-čih* expresses one's reluctant decision or acceptance of a given situation. In (15), the first *-čih* conveys that the subject has already taken the first sip of the alcohol and thus has no choice but to keep drinking it. The underlying instruction here could be that “*you have already touched alcohol and now cannot get back to your original status in which you had not touched alcohol, so just take some more drinks, complying with your present status of having started drinking*”. The absence of *-čih* would make the sentence sound weird or at least less natural than the case with *-čih* present.

- (15) Uu-čih-san yum čin, bas neg comoo uu-čih!
 drink-čih-PST CONJ one more cup drink-čih
 ‘Since you have taken a drink of it, take one more drink’

Reluctant decision is also expressed in (7), in which the subject has reluctantly decided to keep waiting.

Second, *-čih* expresses a regretful feeling. In (16), the speaker regrets that he spent too much on the coat. The adverbial *dendüü ünetei-geer* ‘at too high a price’ contributes information that the speaker is not happy that he bought the coat at that price.

- (16) Bi en deel-ig dendüü ünetei-geer ab-čih-žee.
 I this coat-ACC at too high a price buy-čih-PST
 ‘(I regret that) I bought this coat at too high a price.’

Third, *-čih* expresses disappointment, where the event is reported with an undesirable flavor. In (17), the fact that we were defeated goes against the speaker's wish that we had not been defeated. This is the speaker's presupposition, where *-čih* is used to express his disappointment at the result of the race.

- (17) Bid basal diile-gd-čih-žee.
 we again defeat-PS-čih-PST
 ‘We were defeated again.’

⁹ Modal meanings and use of *-čih* were reported by Otgonsüren (1982), Chuluu (1998) and Brosig (2014) explicitly or non-explicitly. Otgonsüren (1982: 124) states that *-čih* can express the speaker's emotion. One example he provided is *hoyuulaa žaahan uu-čih-ya*, in which *uu-čih-ya* ‘let us drink’ does not express the thorough completion of the event but the speaker's feeling like “*hoyuulaa uuž orhi-yo, yaadag yum be*”, which translates as “let us just drink, do not care what will happen”. Chuluu (1998: 179) writes, “... *-čih* more likely involves a modal meaning rather than a pure aspect. Brosig (2014) discusses *-čih* as a marker of dissatisfaction, which is not an aspectual property.

3 Deriving aspectual and modal meanings of *-čih*

In this section, I show that the aspectual and modal meanings of *-čih* are derived from uncancelledness of events in a way or another and analyze uncancelledness as an aspectual primitive.

3.1 Uncancelledness as an aspectual primitive

How does uncancelledness lead to aspectuality and modality? The key to this question is to explicate the relation between uncancelledness of a given event and R-state (for “resultant state”) that arises from it. R-state is concerned particularly with the perfect. For the perfect, Parsons (1990: 234) observes that “for every event that culminates, there is a corresponding state that holds forever after,” which he labels “R-state”. R-state, unlike T-state (for “target state”), which may or may not last for a long time, cannot cease at some later time because the event occurred and is now over, which is of course something that can never be undone by its very nature. It then follows that R-state and uncancelledness of an action are naturally correlated. Given that R-state is dependent on an occurred action or an action to occur and uncancelledness is the very property of that action per se, it is reasonable to say that R-state has uncancelledness of the action as its source. That is, uncancelledness, being more basic, gives rise to R-state.

With this said, uncancelledness can be formulated as a sort of aspectual primitive. As an aspectual primitive, uncancelledness is understood as the very basic semantic segment that serves to produce one or more semantic concepts. As such, it must play a role somewhere in the domain of, most probably, aspect (and modality as well, as we will see).¹⁰

3.2 Deriving aspectual meanings of *-čih*

Now we look at how uncancelledness plays a role in deriving the aspectual meanings of *-čih*, whose proper function is to mark (thorough) completion of events.¹¹ The notions of completion and uncancelledness are related under an instruction such as

¹⁰ This serves as the first clue for us to identify this notion as the basis of aspectual and modal meanings of *-čih* and arguably of any other cognate forms in any other languages.

¹¹ Brosig (2014) takes what he calls dissatisfaction as the central notion of *-čih*. However, dissatisfaction as a general term is misleading on the one hand, and this notion fails to cover all the meanings of *-čih* on the other, although admittedly what Brosig conveys using this term is largely on the right track.

“To say that an action cannot be cancelled is to say that it is ‘completed thoroughly’”. It is not difficult to see that uncanceledness of an action takes shape by means of having its completion intensified.¹² If an action is not subject to cancellation, its completion is subject to intensification, hence the emphatic value, for example, of *-čih*. Bybee et al. (1994: 57) states that there is a certain emphasis inherent in the notion of having brought an action to a thorough conclusion.¹³

On the other hand, R-state adhering to uncanceledness arises in connection with intensity of completion, which, informally, gives the following instruction: *The action is completed thoroughly and thus cannot be cancelled; the present R-state cannot cease, so stay in that state*. For example, in (15), repeated as (18), we see that the subject is in R-state of having taken the first sip, which is the reason for why the speaker is suggesting the addressee to take one more drink under an instruction: “you have already drunken alcohol and taking one more drink cannot return you to your original status in which you had not drunken alcohol”.

- (18) Uu-čih-san yum čin, bas neg comoo uu-čih!
 drink-čih-PST CONJ one more cup drink-čih
 ‘Since you have taken a drink of it, take one more drink’

As indicated by (18), *-čih* marks the boundary between the inception of the event and R-state resulting from it. That is, for example, once the commencement of the drinking event in (18) or the waiting event in (7), repeated as (19), has completed, corresponding R-states arise, which is a situation the speaker or the subject is reluctant to accept.

- (19) Hüle-čih-sen yum čin, hüle-h I hereg.
 wait-čih-PST CONJ wait-INF have no choice but
 ‘Since I started waiting, I have no choice but to wait.’

In (20), Baatar won the race, giving rise to R-state of having won the race, and so it cannot turn out to be that he has not won yet.

- (20) #Öčigdör-in möriičöön-d Baatar yal-čih-žee. Getel dahin
 yesterday’s race-DAT Baatar win-čih-PST however again
 todol-san-d, harin yal-san-güi yum bai-na.
 as a result of testing instead win-PST-NEG MD be-NPST
 ‘Baatar won yesterday’s race. But he has not won, as a result of retesting.’

¹² Studies such as Sanzheyev (1973: 84, 88), Byambasan et al. (1987: 76), Svantesson (1991: 198), Hashimoto (1993: 5), Sanders and Bat-Ireedui (1999: 87), and Janhunen (2003: 168, 2012: 146) discuss the intensifying function of *-čih*.

¹³ See Bybee et al. (1994: 57ff) for detailed discussion on the relation between completion, affect-edness and intensity, which is instantiated cross-linguistically.

From R-state, the resultativity meaning of *-čih* also follows. Resultativity is a matter of T-state, which currently holds for the participant(s) of the event. Given that R-state collapses necessarily into T-state because they both are along the axis of time, sharing the same temporal dimension, R-state also leads to resultativity. R-state comes about as a result of a past action. That is, once an action takes place, it cannot be cancelled and there is a corresponding R-state: the action is in a state of having occurred. If an action that has passed is never cancelled, it is the case that it has finished *completely/thoroughly*. *-čih* marks such completeness, which leads to resultivity.

In conclusion, *-čih* encodes the notion of uncancelledness, and marks the (thorough) completion of events or of the commencement of events. Uncancelledness gives rise to R-state and therefore is the source of the aspectual meanings of *-čih*.

3.3 Deriving modal meanings of *-čih*

Modal meanings of *-čih*, closely related to its aspectual meanings, also follow from uncancelledness and R-state. Importantly, R-state, arising from an uncancellable action, does not comply with the speaker's or the subject's will. That is, the speaker or the subject is unwilling to accept a given R-state. As such, there is speaker's presupposition in which a contradiction arises between the asserted proposition and an alternative. The asserted proposition goes against the speaker's or the subject's will while the alternative proposition complies with it.

For example, in (18), the speaker presupposes a counterfactual situation in which the first sip of the alcohol did not take place. The action of taking a sip has already occurred and is impossible to be cancelled, which is against the speaker's or the subject's will. It is important to note that *-čih* triggers the presupposition in which a contradiction arises between the asserted proposition of taking the first sip and the counterfactual proposition.

The same holds true in (19), in which the speaker is **reluctant** or unhappy to keep waiting. Since she/he has started waiting, which is a situation that cannot be cancelled, the speaker has reluctantly decided to keep doing it. Again, *-čih* triggers the presupposition in which the actual event of waiting is contradictory with the counterfactual situation in which the waiting did not take place.

In (16), you bought a coat at a no-return shop but now you **regret** it because you now feel that it cost too much. You cannot but accept the situation that you cannot make a return. The buying action has taken place and cannot be cancelled, and you now have the coat as a result of having bought it, which is not what you wish to have happened. A contradiction thus arises between the reported buying event and a presupposed proposition that the buying event does not take place.

In (17), the speaker presupposes a situation in which the result of the race contradicts his expectation.

This modal semantics can be represented informally as follows. In (21), what the speaker chose to assert is [do x], which contradicts [not-do x] in the alternative set.¹⁴

- (21) a. $p (= [\text{not-do } x]) \wedge \neg q (= [\text{do } x])$
 b. I did not buy the coat. I bought the coat.

4 Uncancelledness, resultant state and the perfect

Two distinct states result from the perfect (Parsons 1990: 234–235). T-state is purely aspectual and R-state is temporal-aspectual. What is intrinsically relevant with uncancelledness of an event is R-state, not T-state, although both commence when the event is over (Higginbotham 2008: 175). To give a paradigm illustration for T-state of (22), we obtain (23), where for some state s , s is a state of having the coat, John is the theme of s , s commences when e is over (as represented by the notation R (for “relation”)),¹⁵ and e is a buying, for which John is the agent and the coat is the theme.

- (22) John has bought a coat.
 T-state: John has a coat. (John is in a state of having a coat.)
 R-state: John’s buying a coat has occurred. (John’s buying a coat is in a state of having occurred)

- (23) $(\exists s)[\text{having-coat}(s) \ \& \ \text{Theme}(s, \text{John}) \ \& \ R(s, e) \ \& \ \text{buy}(\text{John}, \text{coat}, e)]$

The interpretation of R-state of (22) is given in (24), where for some state s , s is a state of having-occurred, e is the theme of s , s commences when e is over, and e is a buying, for which John is the agent and the coat is the theme.

- (24) $(\exists s)[\text{having-occurred}(s) \ \& \ \text{Theme}(s, e) \ \& \ R(s, e) \ \& \ \text{buy}(\text{John}, \text{coat}, e)]$

Suppose that John managed to return the coat. In (25), the first conjunct is the perfect but a simple past. This is because T-state resulting from the buying event was terminated by the subsequent culmination of the returning event, complying with Higginbotham’s (2008) conclusion that the (English) perfect is purely aspectual.

¹⁴ A formal treatment in terms of, for example, Rooth’s (2016) alternative semantics, is indicated. Space limitation does not allow a detailed discussion here, however.

¹⁵ I follow Higginbotham (2008: 174) in assuming that the head expresses a relation $R(\text{state}, \text{event})$, his $R(e', e)$, satisfied by an ordered pair of a state and an event such that the state is a result of the event.

(25) John bought the coat and has returned it now.

As shown in (24), s^1 (T-state, i.e. having the coat) lasts only until e^2 occurs. Since John is the agent of e^2 , he is the theme of s^2 . Once e^2 culminates, it terminates s^1 , which remains theme-less because John has been extracted from it to become the theme of s^2 . That is, the original T-state has been tampered by extraction of its theme. It then follows that the length of T-state is dependent on a new event.

(26) $(\exists s)[\text{having-coat}(s^1) \ \& \ \text{Theme}(s^1, \varphi) \ \& \ R(s^1, e^1) \ \& \ \text{buy}(\text{John, coat, } e^1)] \ \& \\ (\exists s)[\text{having-no-coat}(s^2) \ \& \ \text{Theme}(s^2, \text{John}) \ \& \ R(s^2, e^2) \ \& \ \text{return}(\text{John, coat, } e^2)]$

By contrast, as shown in (27), for R-state, s^1 lasts forever because it is not subject to tampering. Notice that when a new event e^2 , namely returning-of-the-coat, occurs subsequently, the corresponding R-state s^2 , namely having-occurred, has e^2 , not e^1 , as its theme, with e^1 remaining where it was.

(27) $(\exists s)[\text{having-occurred}(s^1) \ \& \ \text{Theme}(s^1, e^1) \ \& \ R(s^1, e^1) \ \& \ \text{buy}(\text{John, coat, } e^1)] \ \& \\ (\exists s)[\text{having-occurred}(s^2) \ \& \ \text{Theme}(s^2, e^2) \ \& \ R(s^2, e^2) \ \& \ \text{return}(\text{John, coat, } e^2)]$

As such, the no-tampering property of R-state intrinsically relates to uncanceledness of e .

5 Conclusion and discussion

The following issues are noted in this concluding section. To summarize what has been discussed in preceding sections: **Firstly**, uncanceledness is the very basic notion underlying perfect-related aspectuality and (un)willingness-related related modality and therefore is a semantic primitive. The overlap between R-state, arising from uncanceledness, and T-state, arising from completion of event, in the same temporal axis leads to aspectuality on the one hand and the noncompliance of R-state with the speaker's will leads to modality on the other hand; **secondly**, the perfect is not only aspectual but also modal. Similarly, the simple past is not purely temporal but rather temporal, aspectual, and modal. The underlying notion that connects aspectuality and modality (and perhaps temporality as well) is uncanceledness; **thirdly**, contradiction-involving presupposition may be triggered by uncanceledness, which falls in alternative semantics formulated by, for example, Rooth (2016).

Cross-linguistically, *tesimau* in Japanese and *le* in Chinese, both characterized as aspectual morphemes, have been observed to have a modal sense of contrary-to-expectation (Ishikawa 1992; Soh 2009). Interestingly, *tesimau*, which Ishikawa (1992) relates to unrecoverability of a pre-event situation, originates as a content lexeme meaning "to finish", and *le* originates as the full-fledged content lexeme *liǎo*, which

also means “to finish”. Bybee et al. (1994: 57–61) reported a range of morphemes or constructions in some other languages and related them to notions such as (thorough) completion of events, total affectedness of the patient, and emphatic value. As implied by the analysis presented in this paper, these notions may all be underlain by a common basic notion, namely uncanceledness.

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